

IMO Training – Number Theory

Elementary Number Theory: Miscellaneous Facts

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Some Advanced Theorems

Theorem. (Fermat's Two Square Theorem) Any prime p of the form $4k + 1$ is a sum of two squares.

Theorem. (Fermat's Last Theorem) If $x, y, z, n \in \mathbb{Z}$, $n \geq 3$ and $x^n + y^n = z^n$, then $xyz = 0$.

Theorem. (Dirichlet) If $(a, b) = 1$, then there are infinitely many primes of the form $an + b$.

Theorem. (Catalan) The only integral solution to the equation $a^b - c^d = 1$ with $a, b, c, d > 1$ is $(a, b, c, d) = (3, 2, 2, 3)$.

Theorem. (Fibonacci Power) 1, 8, and 144 are the only perfect powers in the Fibonacci sequence.

Example. Show that $a^2 + b$ and $b^2 + a$ cannot be both perfect squares.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 - y^3 = xy + 61$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - y! = 2001$.

Example. For $m, n \in \mathbb{N}$, find the minimum value of $|12^m - 5^n|$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^4 = 19^{19}$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^5 = y^2 + 4$.

Example. Find all $k \in \mathbb{N}$ s.t. $\exists a, b \in \mathbb{N}$ with $a^2 + b^2 = kab$.

Example. If $a, b \in \mathbb{N}$ and $b^2 + ab + 1 \mid a^2 + ab + 1$, prove that $a = b$.

Example. Show that for $a, b, c \in \mathbb{N}$ with $a^2 + b^2 = c^2$, $\exists m, n \in \mathbb{Z}$ s.t.

$$(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2).$$

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^2 - 2y^2 = 1$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $3x^2 + 1 = 4y^3$.

Exercises

1. (APMO 2011) Let a, b, c be positive integers. Prove that it is impossible to have all of the three numbers $a^2 + b + c, b^2 + a + c, c^2 + a + b$ to be perfect squares.
2. Find all $x, y \in \mathbb{N}$ s.t. $7^x = 3^y + 4$.
3. Find all $x, y \in \mathbb{N}$ s.t. $2^x + 3 = 11^y$.
4. (Korea 1993) An integer which is the area of a right-angled triangle with integer sides is called *Pythagorean*. Prove that for every positive integer $n > 12$ there is a Pythagorean number between n and $2n$.

Extra Space